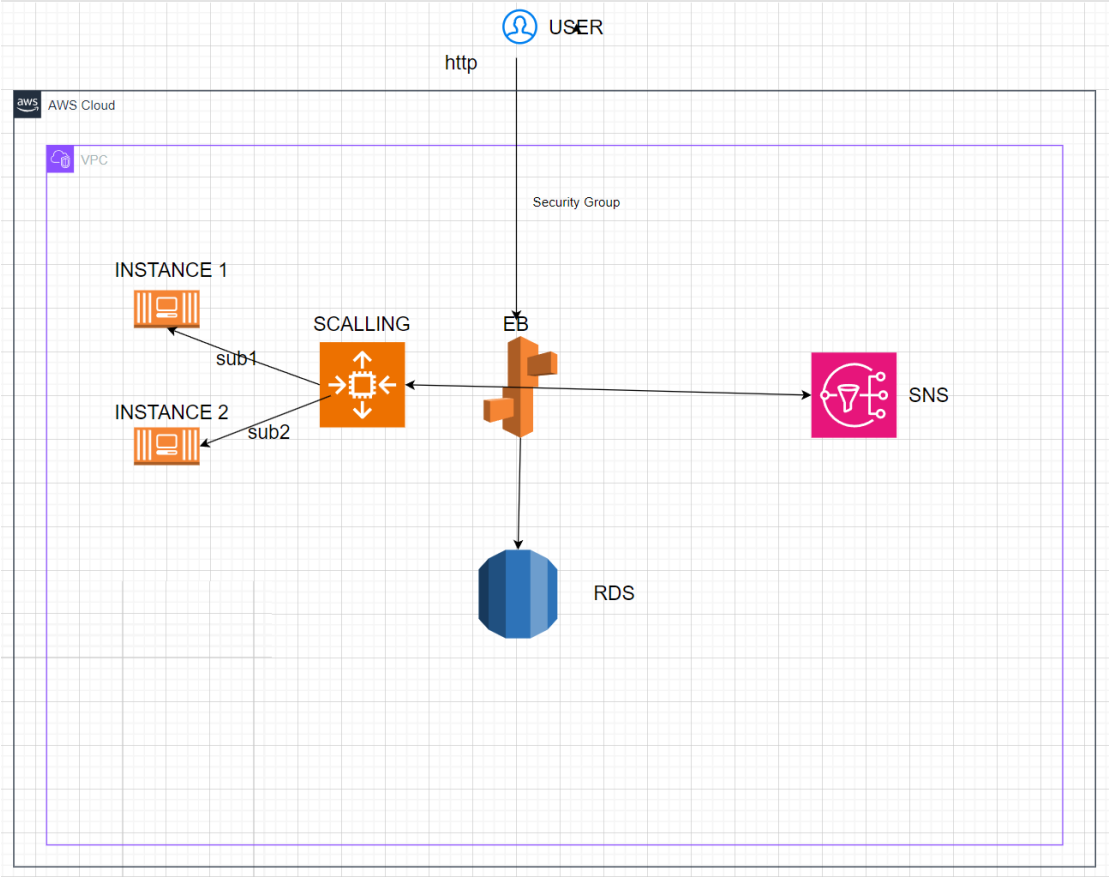
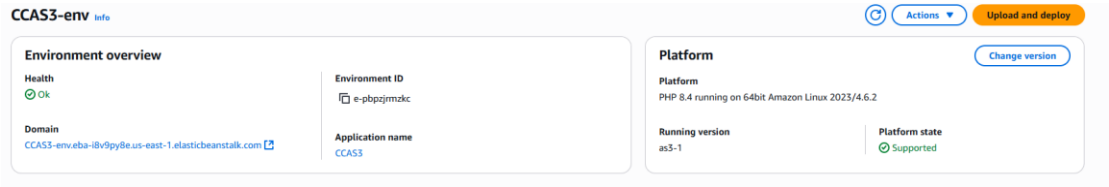
Architecture Diagram



AWS EB (include green status)

In order to meet the requirements of the enterprise, I created EB and made it meet the requirements.



Configure environment Info

Environment tier Info

Amazon Elastic Beanstalk has two types of environment tiers to support different types of web applications.

- (●) Web server environment
  Run a website, web application, or web API that serves HTTP requests. Learn more
- ( ) Worker environment
  Run a worker application that processes long-running workloads on demand or performs tasks on a schedule. Learn more

Application information Info

Application name

CCAS3

Maximum length of 100 characters.

▶ **Application tags (optional)**

Environment information Info

Choose the name, subdomain and description for your environment. These cannot be changed later.

Environment name

CCAS3-env

Must be from 4 to 40 characters in length. The name can contain only letters, numbers, and hyphens. It can't start or end with a hyphen. This name must be unique within a region in your account.

Domain

Leave blank for autogenerated value .us-east-1.elasticbeanstalk.com Check availability

Environment description

Platform Info

Platform type

- (●) Managed platform
  Platforms published and maintained by Amazon Elastic Beanstalk. Learn more
- ( ) Custom platform
  Platforms created and owned by you. This option is unavailable if you have no platforms.

Platform

PHP

Platform

PHP

Platform branch

PHP 8.4 running on 64bit Amazon Linux 2023

Platform version

4.6.2 (Recommended)

Application code Info

- ( ) Sample application
- ( ) Existing version
  Application versions that you have uploaded.
- (●) Upload your code
  Upload a source bundle from your computer or copy one from Amazon S3.

Version label

Unique name for this version of your application code.

as3

Source code origin. Maximum size 500 MB

- (●) Local file

  Upload application

  Choose file

  ✓ File name: **index.php**

  File must be less than 500MB max file size
- ( ) Public S3 URL

Presets Info

Start from a preset that matches your use case or choose custom configuration to unset recommended values and use the service's default values.

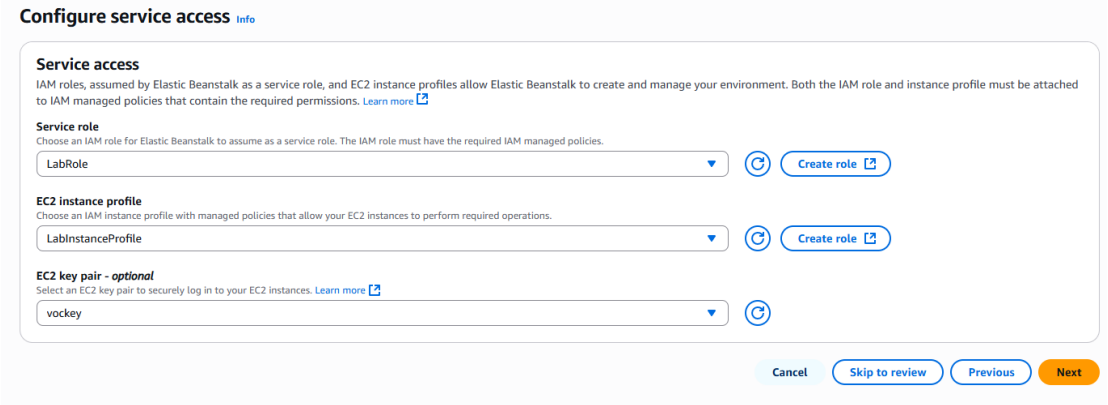
Configuration presets

- ( ) Single instance (free tier eligible)
- ( ) Single instance (using spot instance)
- ( ) High availability
- ( ) High availability (using spot and on-demand instances)
- (●) Custom configuration

Cancel Next

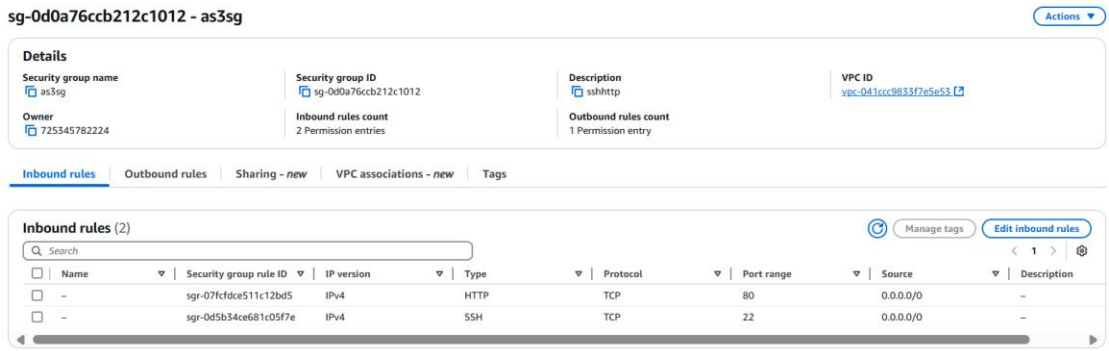
key pairs

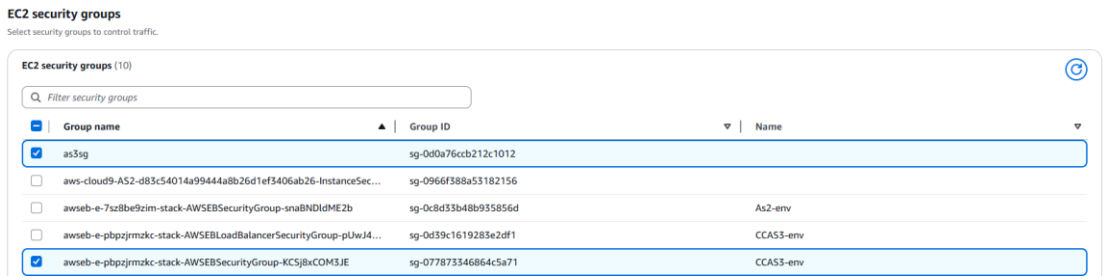
All the instance use the same key vockey.



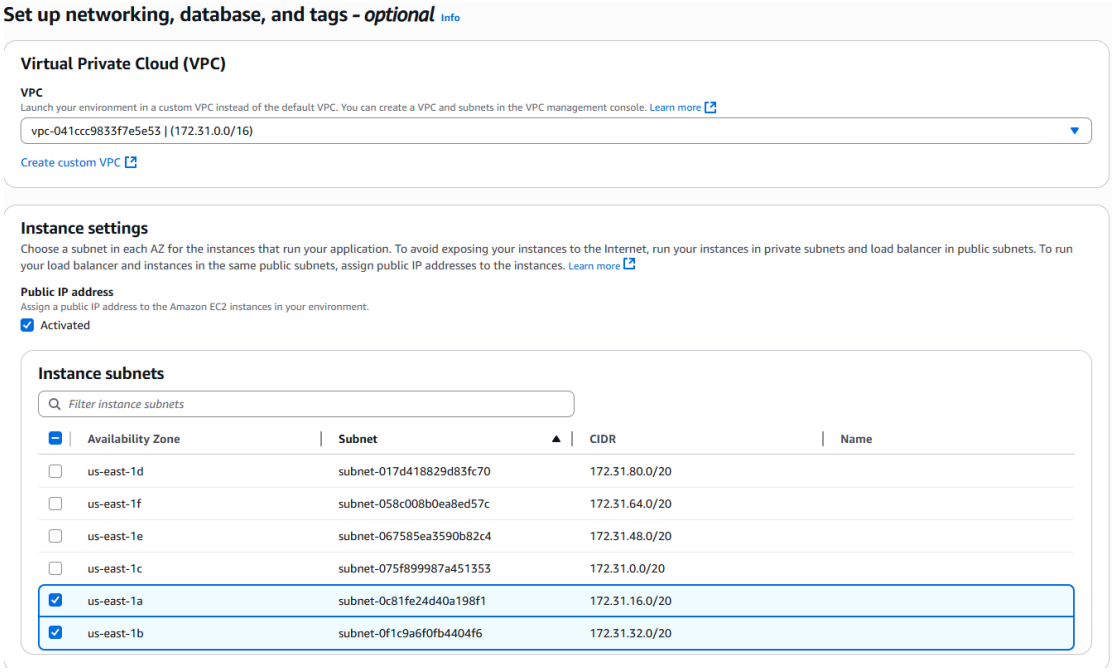
Custom Security groups

I create security group to make HTTP and SSH are allowed.





VPC with at least two subnets in different Availability zones



Set up networking, database, and tags - *optional* Info

Virtual Private Cloud (VPC)

VPC

Launch your environment in a custom VPC instead of the default VPC. You can create a VPC and subnets in the VPC management console. Learn more

vpc-041ccc9833f7e5e53 | (172.31.0.0/16)

Create custom VPC

Instance settings

Choose a subnet in each AZ for the instances that run your application. To avoid exposing your instances to the Internet, run your instances in private subnets and load balancer in public subnets. To run your load balancer and instances in the same public subnets, assign public IP addresses to the instances. Learn more

Public IP address

Assign a public IP address to the Amazon EC2 instances in your environment.

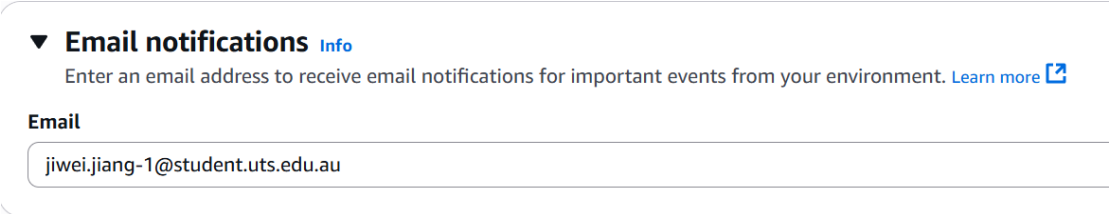
☑ Activated

Instance subnets

Filter instance subnets

Availability Zone	Subnet	CIDR	Name
us-east-1d	subnet-017d418829d83fc70	172.31.80.0/20	
us-east-1f	subnet-058c008b0ea8ed57c	172.31.64.0/20	
us-east-1e	subnet-067585ea3590b82c4	172.31.48.0/20	
us-east-1c	subnet-075f899987a451353	172.31.0.0/20	
us-east-1a	subnet-0c81fe24d40a198f1	172.31.16.0/20	
us-east-1b	subnet-0f1c9a6f0fb4404f6	172.31.32.0/20	

Email notification



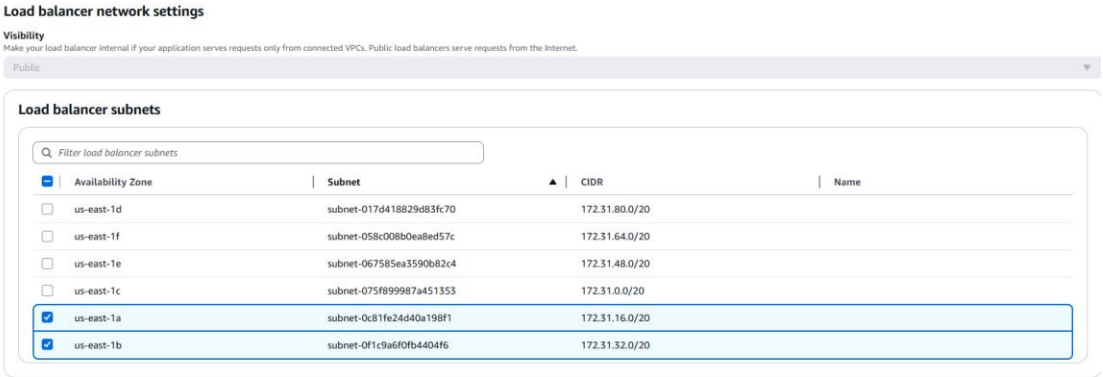
Email notifications Info

Enter an email address to receive email notifications for important events from your environment. Learn more

Email

jiwei.jiang-1@student.uts.edu.au

Load Balancer



Load balancer network settings

Visibility

Make your load balancer internal if your application serves requests only from connected VPCs. Public load balancers serve requests from the Internet.

Public

Load balancer subnets

Filter load balancer subnets

Availability Zone	Subnet	CIDR	Name
us-east-1d	subnet-017d418829d83fc70	172.31.80.0/20	
us-east-1f	subnet-058c008b0ea8ed57c	172.31.64.0/20	
us-east-1e	subnet-067585ea3590b82c4	172.31.48.0/20	
us-east-1c	subnet-075f899987a451353	172.31.0.0/20	
us-east-1a	subnet-0c81fe24d40a198f1	172.31.16.0/20	
us-east-1b	subnet-0f1c9a6f0fb4404f6	172.31.32.0/20	

Database and RDS

Database Info

Integrate an RDS SQL database with your environment. Learn more

Database subnets

If your Elastic Beanstalk environment is attached to an Amazon RDS, choose subnets for your database instances. Learn more

Choose database subnets (6)

Filter database subnets

	Availability Zone	Subnet	CIDR	Name
☐	us-east-1d	subnet-017d418829d83fc70	172.31.80.0/20	
☐	us-east-1f	subnet-058c008b0ea8ed57c	172.31.64.0/20	
☐	us-east-1e	subnet-067585ea3590b82c4	172.31.48.0/20	
☐	us-east-1c	subnet-075f899987a451353	172.31.0.0/20	
☑	us-east-1a	subnet-0c81fe24d40a198f1	172.31.16.0/20	
☑	us-east-1b	subnet-0f1c9a6f0fb4404f6	172.31.32.0/20	

Enable database

Restore a snapshot - *optional*

Restore an existing snapshot from a previously used database.

Snapshot

None

Database settings

Choose an engine and instance type for your environment's database.

Engine

mysql

Engine version

8.0.41

Instance class

db.t3.small

Storage

Choose a number between 5 GB and 1024 GB.

5 GB

Username

as3db

Database settings

Choose an engine and instance type for your environment's database.

Engine

mysql

Engine version

8.0.41

Instance class

db.t3.small

Storage

Choose a number between 5 GB and 1024 GB.

5 GB

Username

as3db

Password

••••••••••

Availability

High (Multi-AZ)

Database deletion policy

This policy applies when you decouple a database or terminate the environment coupled to it.

- ○ Create snapshot
  Elastic Beanstalk saves a snapshot of the database and then deletes it. You can restore a database from a snapshot when you add a DB to an Elastic Beanstalk environment or when you create a standalone database. You might incur charges for storing database snapshots.
- ○ Retain
  The decoupled database will remain available and operational external to Elastic Beanstalk.
- ◉ Delete
  Elastic Beanstalk terminates the database. The database will no longer be available.

Tags

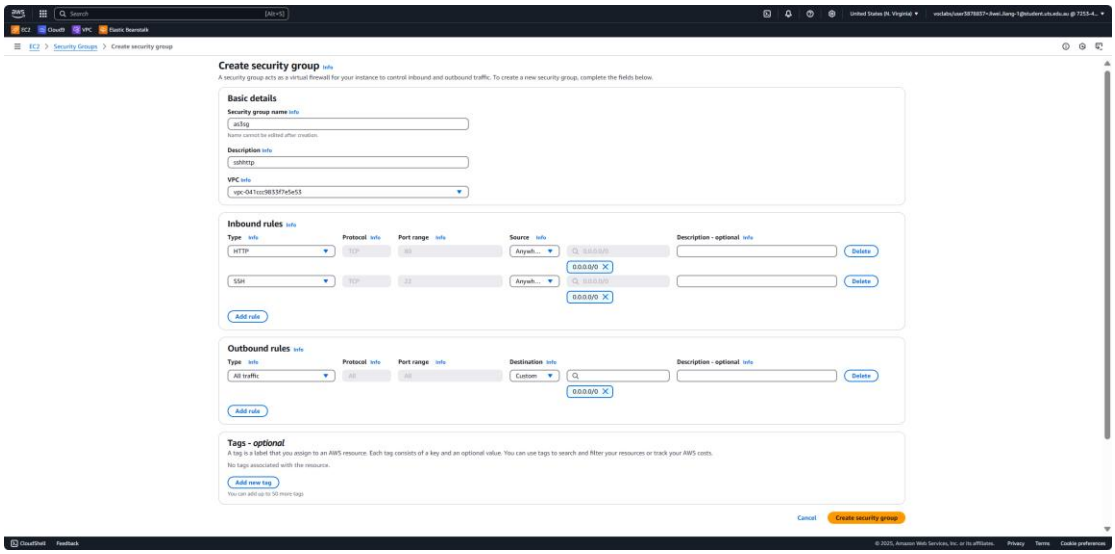
Apply up to 50 tags. You can use tags to group and filter your resources. A tag is a key-value pair. The key must be unique within the resource and is case-sensitive. Learn more

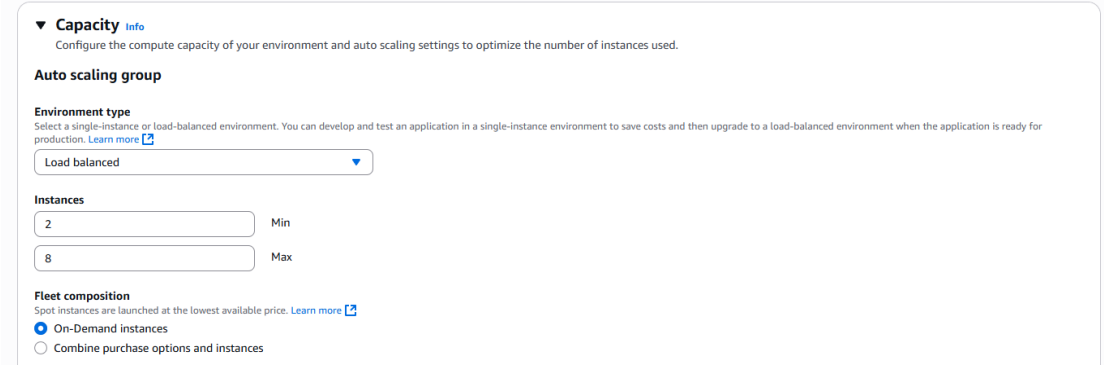
No tags associated with the resource.

Add new tag

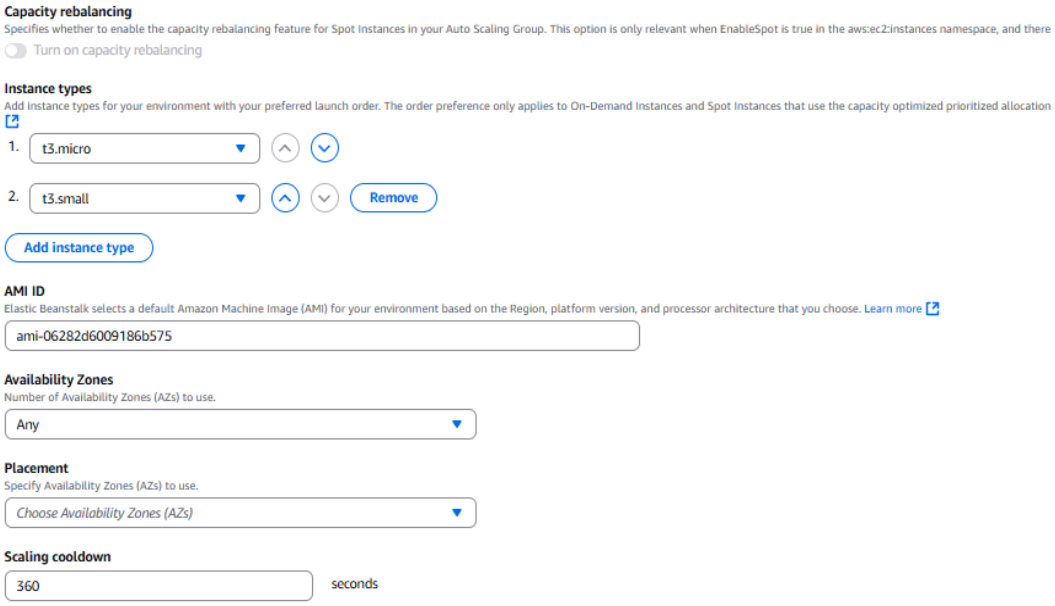
You can add 50 more tags.

Cancel | Skip to review | Previous | Next





I Create new AMI.



Copy Amazon Machine Image (AMI)

Original AMI ID

ami-06282d6009186b575

AMI copy name

ccas3-php-ami-v1

AMI copy description

[Copied ami-06282d6009186b575 from us-east-1] ccas3-php-ami-v1

Auto Scaling (Set scaling triggers on network output traffic with an upper threshold of 60% and a lower threshold of 30%.)

Metric

Change the metric that is monitored to determine if the environment's capacity is too low or too high.

NetworkOut

Statistic

Choose how the metric is interpreted.

Average

Unit

Percent

Period

The period between metric evaluations.

5 Min

Breach duration

The amount of time a metric can exceed a threshold before triggering a scaling operation.

5 Min

Upper threshold

60

Scale up increment

1 EC2 instances

Lower threshold

30 capacity

Scale down increment

-1 EC2 instances

Time-based scaling

Use the following settings to control time-based scaling actions.

Current status

2 instance(s) in service, Min: 2, Max 8

Time zone

- (●) UTC
- ( ) Local